



AKTU

B.Tech 1st Year



Environment & Ecology

ONE SHOT Revision

Unit-3

Pollution and their Effects

Notes I PYQs



by M S Tomar Sir

Environment and Ecology

AKTU Syllabus : Unit-3

Environment: Pollution and their Effects; Public Health Aspects of Environmental

- Water Pollution
- Air Pollution
- Soil Pollution
- Noise Pollution
- Solid waste management

AKTU : B.Tech (III-Sem)

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Pollution: It is defined as the addition of **undesirable material** into the environment as a result of human activities.

Pollutants: A pollutant may be defined as a physical, chemical or biological substance released into the environment which is directly or indirectly harmful to humans and other living organisms.

Legislations against Pollution (प्रदूषण के खिलाफ कानून)

- The Water (Prevention and Control of Pollution) Act, 1974
- The Air (Prevention and Control of Pollution) Act, 1981
- The Environment Protection Act, 1986
- The Hazardous Waste Management Regulations
- E-Waste (Management and Handling) Rules, 2011
- Plastic waste Management Rules

Classification of Pollutants

1. According to the form in which they persist after release into the environment.

a) Primary pollutants

→ These are persistent in the form in which they are added to the environment

Examples:- → Carbon monoxide(CO) - released from the incomplete combustion of fossil fuels

→ Nitrogen oxides- released by industry and automobiles

→ Sulfur oxides - emitted from burning coal

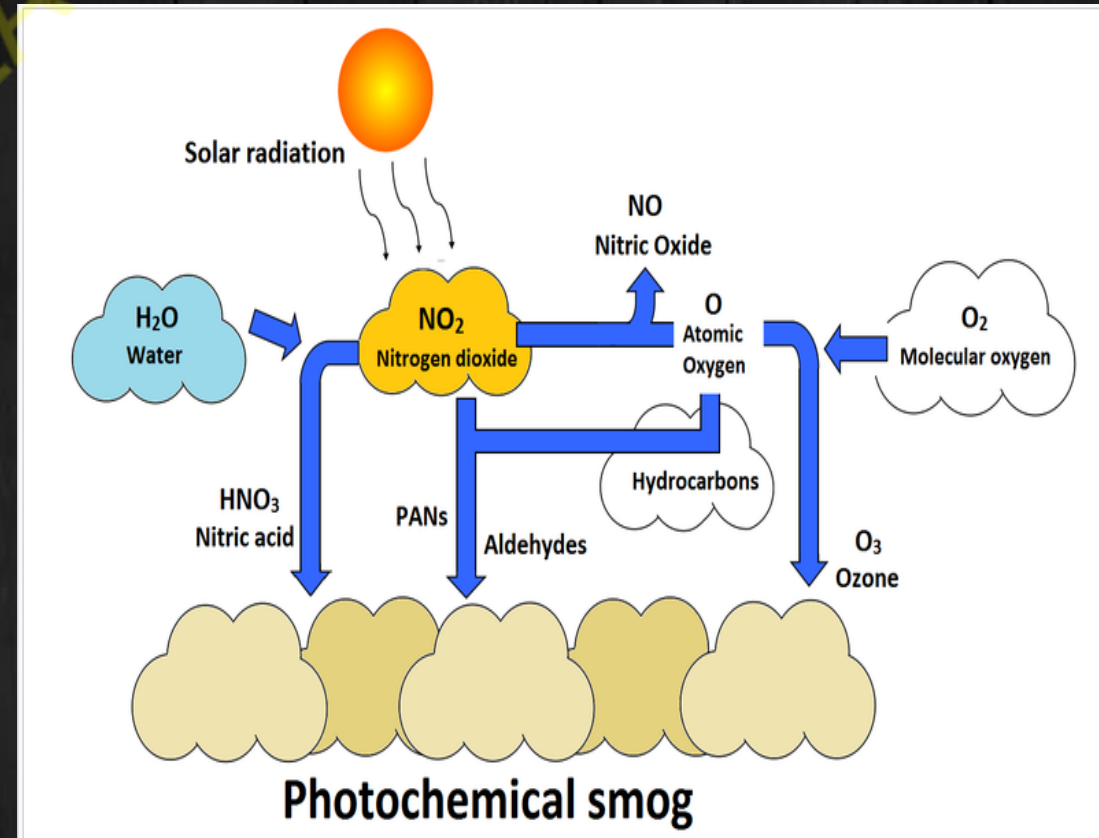
b) Secondary Pollutants

→ These are formed by interaction among the primary pollutants.

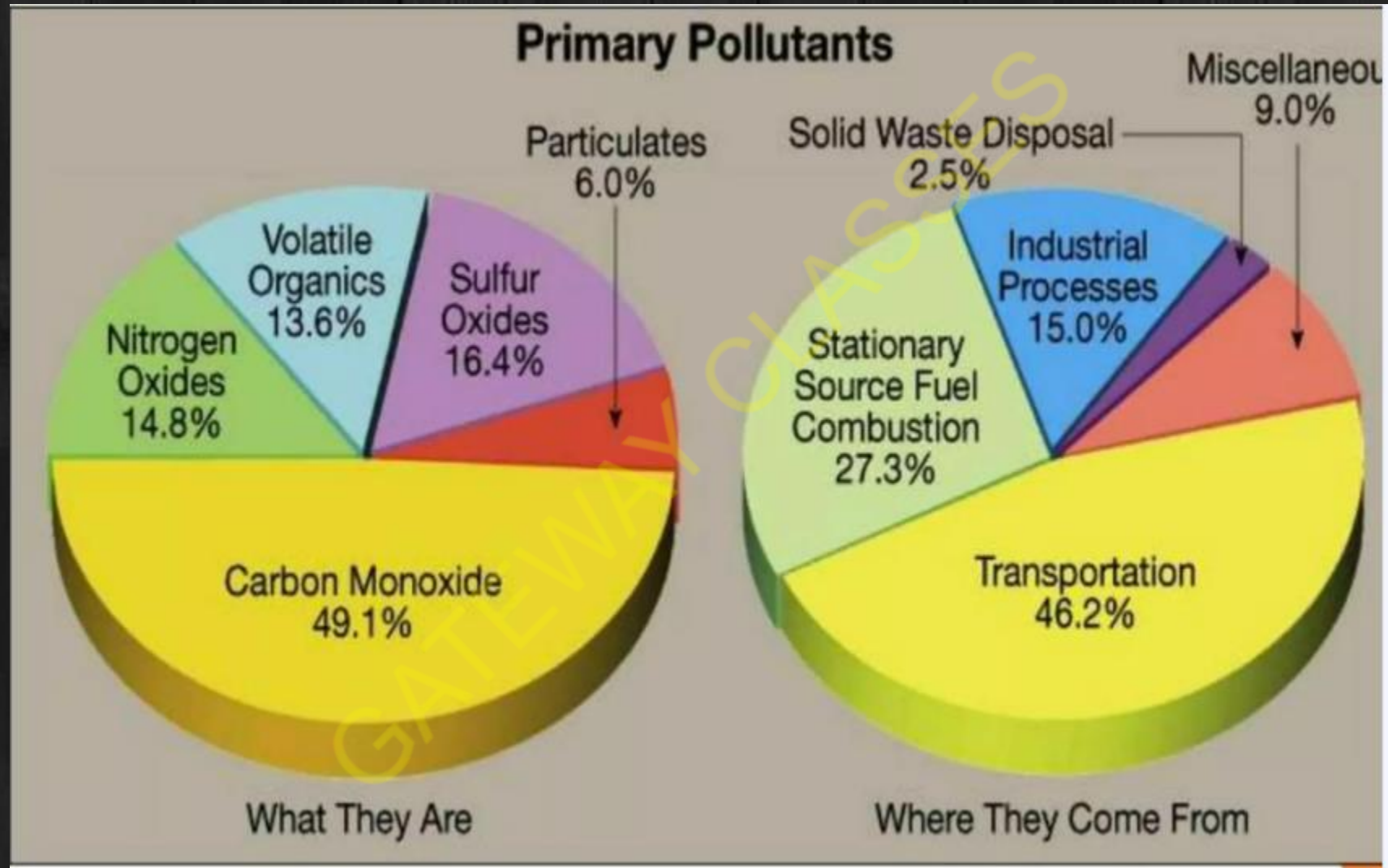
Examples:-

→ Photochemical Smog → $NO_x + VOC \rightleftharpoons PAN + O_3$

→ Ozone



Primary Pollutants



2. According to their existence in nature

- a) **Quantitative Pollutants:** These occur in nature and become pollutants when their concentration reaches beyond a threshold level.

Examples: carbon dioxide, nitrogen oxide etc.

- b) **Qualitative Pollutants:** These do not occur in nature and are man-made.

Examples: fungicides, herbicides, DDT etc.

3. According to their frequency of release

- a) **Chronic Pollutants:** A persistent release of a pollutant at a low concentration resulting in adverse effects on animal or human body with symptoms that develop slowly. Such symptoms do not usually subside when the exposure stops. **Examples:** Second Hand exposure to Cigarette smoke.

- b) **Acute Pollutants (तीव्र प्रदूषक) :** Pollution of significance which occurs suddenly and should take into account worst case discharge scenario from the facility. **Examples:** are larger oil spill due to pipe rupture or a blow out from a well.

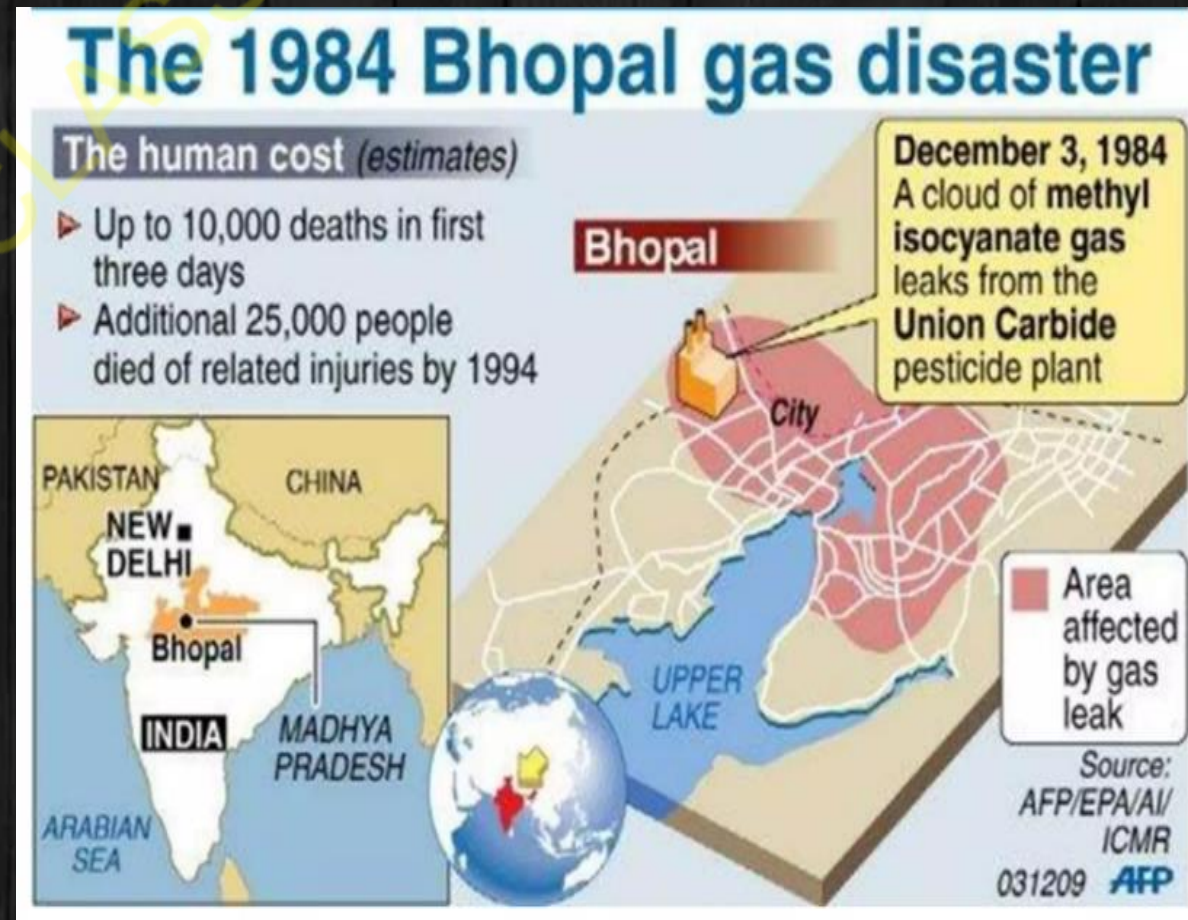
Chronic –

- Long term release of small amount
- Symptoms appear after long term, low level exposure
- Air pollution releasing particulate matter(PM)

Acute –

- Large amounts of pollutant released at one time
- Symptoms appear soon after short, intense exposure
- Bhopal Disaster 1984-Methyl isocyanate

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4. According to their nature of disposal

a) Biodegradable Pollutants:

→ Waste products, which are degraded by microbial action.

Examples: sewage, Food Waste.



b) Non-biodegradable Pollutant (many of them are Persistent Organic Pollutants POPs):

→ Pollutants, which are not decomposed by microbial action.

Examples: PCB, Dioxin, DDT, salts of heavy metals, radioactive substances etc.,)

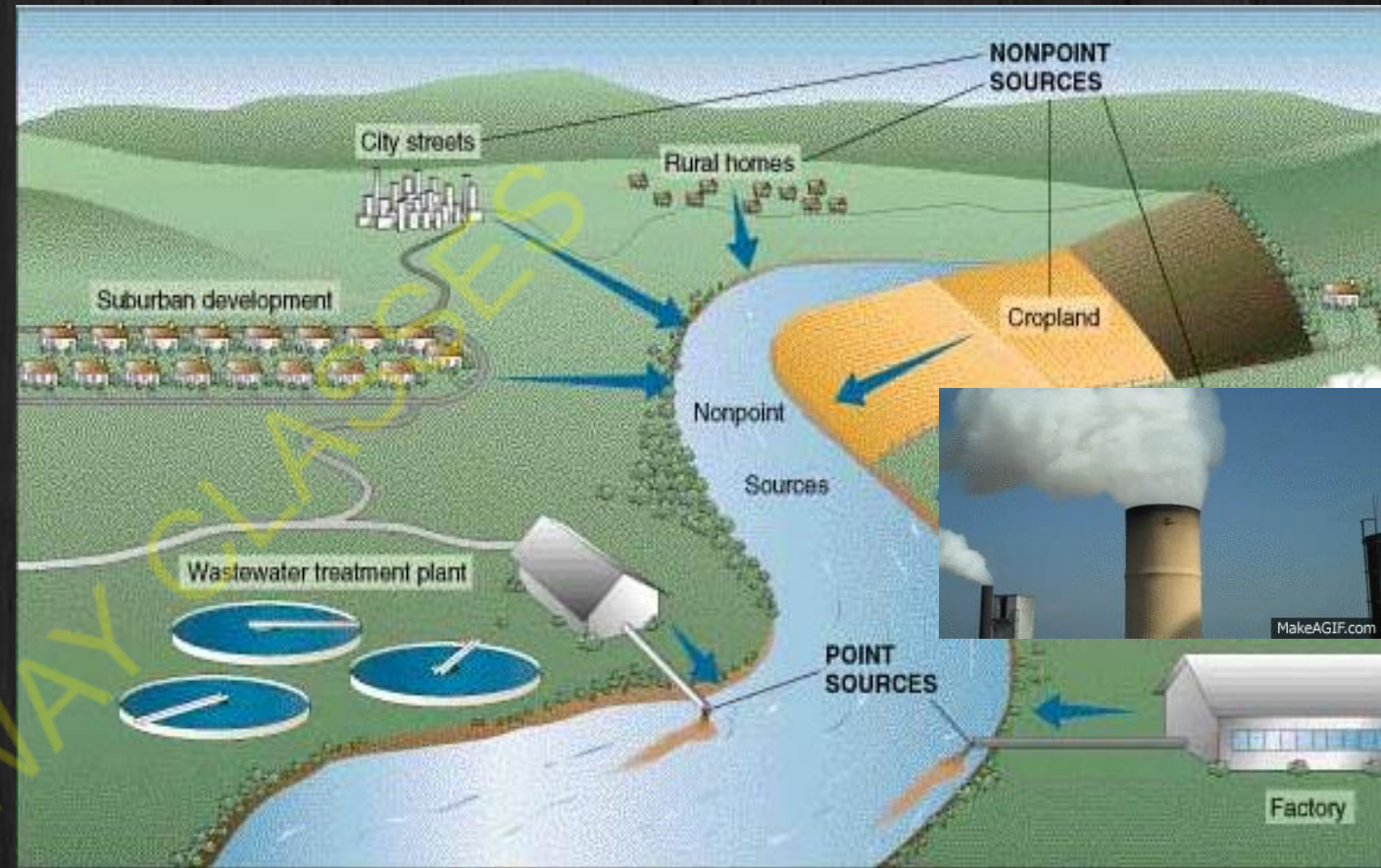


Point source: Pollution that can be traced back to a single origin or source

- Regulation is simpler
- Localized
- Often a single party is responsible
- One jurisdiction

Examples

- Sewage treatment plant discharge
- Drainage pipe
- Leaking storage tank
- Factory chimney

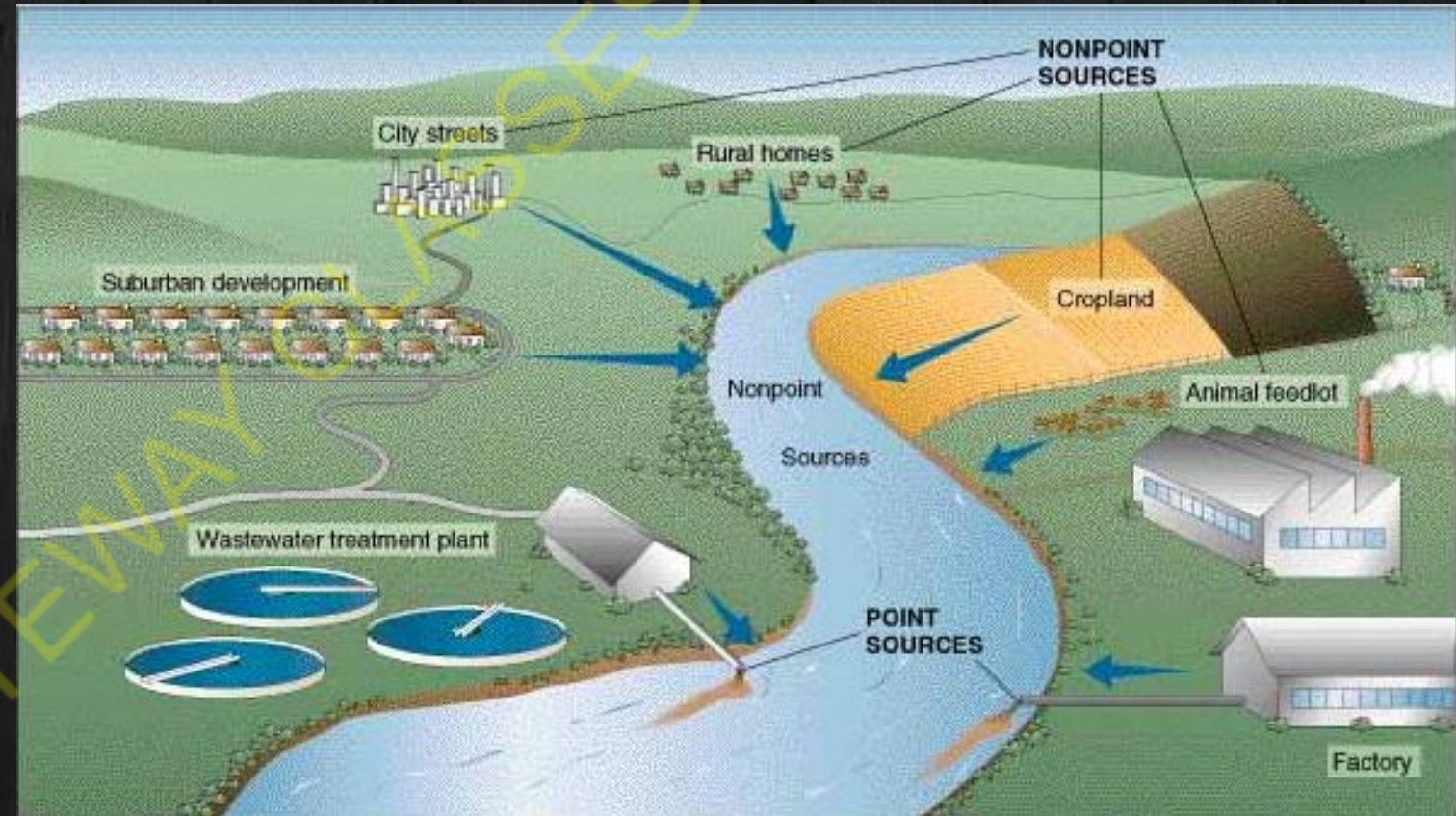


Non-point source: Pollution which cannot be traced back to a single origin or source.

- Regulation is more challenging
- Identify multiple sources
- Who is responsible
- Different legal jurisdictions
- Large scale intervention

Examples

- Agricultural runoff
- Vehicle exhaust
- Industrial emission



Air Pollution

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- Air pollution refers to the contamination of the Earth's atmosphere by harmful substances, including gases, particulates, and biological molecules.
- **The main** culprit for air pollution is the burning of fossil fuels.
- Fossil fuels contain small amounts of nitrogen and sulphur.
- Burning of fossil fuels like coal (thermal power plants) and petroleum release different oxides of nitrogen and sulphur into the atmosphere.
- These gases react with the water vapour present in the atmosphere to form sulphuric acid and nitric acid.
- The acids drop down with rain, making the rain acidic. This is called acid rain.
- Industrial Processes also release various other gasses in atmosphere as well.



Major Air Pollutants and their Sources

1. Carbon monoxide (CO)



→ It is a colourless, odourless gas that is produced by the incomplete burning of carbon-based fuels including petrol, diesel, and wood.

2. Carbon dioxide (CO₂)

→ It is the principle greenhouse gas emitted as a result of human activities such as the burning of coal, oil, and natural gases.

3. Chloroflorocarbons (CFC)

→ These are gases that are released mainly from **air-conditioning systems and refrigeration**.

→ When released into the air, CFCs rise to the stratosphere, where they come in contact with few other gases, which lead to a reduction of the ozone layer that protects the earth from the harmful ultraviolet rays of the sun.

4. Lead

- It is present in petrol, diesel, lead batteries, paints, hair dye products, etc. Lead affects children in particular.
- It can cause nervous system damage and digestive problems and, in some cases, cause cancer.

5. Ozone

- It occurs naturally in the upper layers of the atmosphere. This important gas shields the earth from the harmful ultraviolet rays of the sun.
- However, at the ground level, it is a pollutant with highly toxic effects.
- Vehicles and industries are the major source of ground level ozone emissions.
- Ozone makes our eyes itch, burn, and water. It lowers our resistance to cold and pneumonia.

6. Suspended particulate matter (SPM)

- It consists of solids in the air in the form of smoke, dust, and vapour, that can remain suspended for extended periods and is also the main source of haze which reduces visibility.
- The finer of these particles, when breathed in can lodge in our lungs and cause lung damage and respiratory problems

7. Sulphur dioxide (SO₂)

- It is a gas produced from burning coal, mainly in thermal power plants.
- Some industrial processes, such as the production of paper and the smelting of metals, produce sulphur dioxide.
- It is a major contributor to smog and acid rain.
- Sulfur dioxide can lead to lung diseases.

8. Smog

- Smog has been coined from a combination of the words fog and smoke. Smog is a condition of fog that had smoke in it.
- Photochemical smog (smog) is a term used to describe air pollution that is a result of the interaction of sunlight with certain chemicals in the atmosphere.
- Nitrogen oxides (primarily from vehicle exhaust)
- Volatile Organic Compounds (from paints, solvents, printing inks, petroleum products, vehicles, etc.) interact in the presence of sunlight.

Indoor Air Pollution

- Pollution within a home, or an institution or commercial facility.
- It is the rural areas that face the greatest threat from indoor pollution, where people rely on traditional fuels such as firewood, charcoal, and cow dung for cooking and heating.



In urban areas, exposure to indoor air pollution has increased due to a variety of reasons, such as:

- Construction of more tightly sealed buildings
- Reduced ventilation
- The use of synthetic materials for building and furnishing
- The use of chemical products, pesticides, and household care products.
- Indoor air pollution can begin within the building or drawn in from outdoors.

Prevention and Control of Air Pollution

Control of Indoor Air Pollution

- Use of wood and dung cakes should be replaced by cleaner fuels such as biogas, LPG or electricity.
- The use of solar cookers must be encouraged.
- Charcoal is a comparatively cleaner fuel
- Indoor pollution due to the decay of exposed kitchen waste can be reduced by Covering the waste properly.
- Segregation of waste, pre-treatment at the source, sterilisation of rooms will help.

Control of Industrial Air Pollution

- Use of cleaner fuels such as LNG in power plants, fertiliser plants etc. which is cheaper in addition to being environmentally friendly.
- Ban on Petcoke and increased Coal cess
- Employing environment-friendly industrial processes so that emission of pollutants and hazardous waste is minimized.
- Installing devices which reduce the release of pollutants. Devices like filters, electrostatic precipitators, inertial collectors, scrubbers, gravel bed filters, or dry scrubbers.
- Shift to Renewable Source of Energy like Solar and Wind.

Control of Vehicular Air Pollution

- Earlier lead in the form of tetraethyl lead was added in the petrol to raise octane level for the smooth running of engines. Addition of lead in petrol has been banned to prevent the emission of lead particles.
- CNG in public transport vehicles is made mandatory in cities like Delhi
- Shift to TOD model of urban development based on metro trains
- BS-VI from 1st April 2020

Monitoring of Air Pollution

→ National Air Quality Monitoring Programme Run by CPCB to ascertain the compliance of NAAQS (National Ambient Air Quality Standards)

National Ambient Air Quality Standards (NAAQS) measures the amount of 12 pollutants, which include:

→ Sulphur dioxide (SO_2), nitrogen dioxide (NO_2), PM_{10} , $\text{PM}_{2.5}$, ozone, lead, carbon monoxide (CO), arsenic, nickel, benzene, ammonia, benzopyrene

National Air Quality Index (AQI)

→ Launched by the Environment Ministry in April 2015 under 'Swachh Bharat Abhiyan'

→ The AQI will consider eight pollutants (PM_{10} , $\text{PM}_{2.5}$, NO_2 , SO_2 , CO , O_3 , NH_3 , and Pb).

Fly Ash

- Ash is produced whenever the combustion of solid material takes place.
- Fly ash is a very fine powder generated by the burning of coal in chimneys mainly in thermal power plant.
- It tends to travel far in the air.
- The ash which does not rise is termed as bottom ash.

Fly ash particles are oxide-rich in

- Aluminium silicate (in large amounts)
- Silicon dioxide(SiO_2)
- Calcium Oxide (CaO)
- **Toxic heavy metals** like lead, arsenic, cobalt and copper

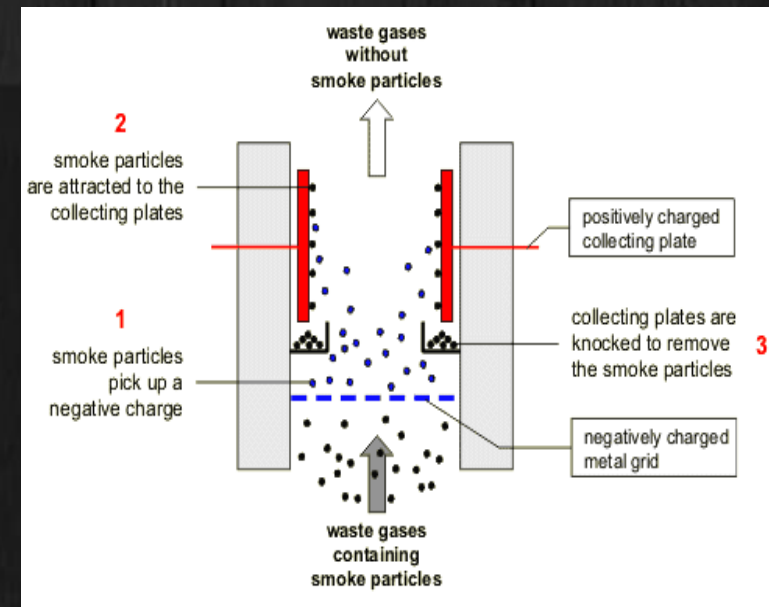
Fly ash is captured by Electrostatic Precipitators

Environmental effects?

- If fly ash which is not captured and disposed off properly, it can pollute air and water considerably.
- It causes respiratory problems.
- Fly ash in the air slowly settles on leaves and crops in areas near to thermal power plants and lowers the plant yield.

Advantages:

- Cement can be replaced by fly ash upto 35%, thus reducing the cost of construction, making roads, etc.
- Fly ash bricks are light in weight and offer high strength and durability.
- Fly ash is a better fill material for roads.
- Fly ash can be used in reclamation of wastelands.
- Abandoned mines can be filled up with fly ash.
- Fly ash can increase the crop yield and it also enhances water holding capacity of the land.



Policy measures of MoEF:

- The Ministry of Environment and Forests vide its notification in 2009, has made it mandatory to use Fly Ash based products in all construction projects, road embankment works and low lying land filling works within 100 km radius of Thermal Power Station.
- To use Fly Ash in mine filling activities within a 50 km radius of Thermal Power Stations.

GATEWAY CLASSES

Water Pollution

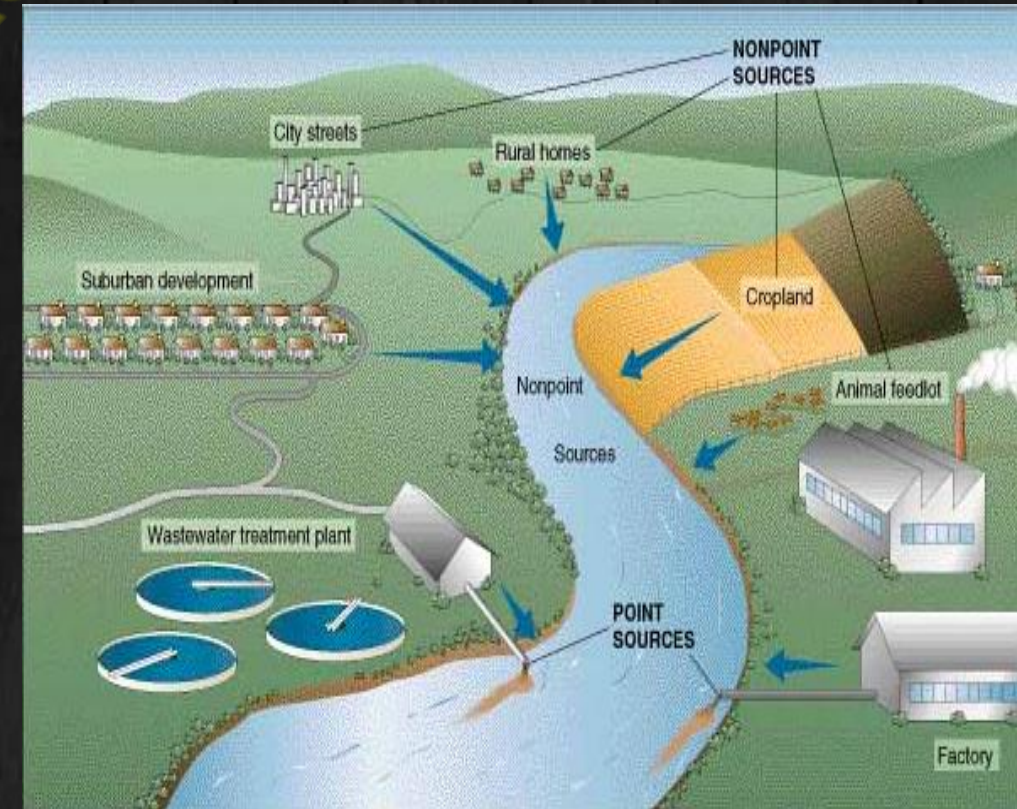
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- Water pollution is the presence of undesirable substances in water such as organic, inorganic, biological, radiological, heat, which degrades the quality of water so that it becomes unfit for use'.
- Natural sources of pollution of water are soil erosion, leaching of minerals from rocks and decaying of organic matter.

Point and non-point sources of pollution

Point Sources: When pollutants are discharged from a specific location such as a drain or pipe carrying industrial effluents discharged directly into a water body it represents point source pollution.

Non-point sources: include discharge of pollutants from diffused sources or from a larger area such as runoff from agricultural fields, grazing lands, construction sites, abandoned mines and pits, moving vehicles.



BOD, DO, COD

- Water pollution by organic wastes is measured in terms of **Biochemical Oxygen Demand (BOD)**.
- BOD is the amount of dissolved oxygen needed by bacteria to decompose the organic wastes present in water. It is expressed in milligrams of oxygen per litre of water.
- The higher value of BOD indicates low DO content of water. Since BOD is limited to biodegradable materials only. Therefore, it is not a reliable method of measuring pollution load in water.

BOD Level in mg/liter	Water Quality
1-2	Very Good: There will not be much organic matter present in the water supply
3.5	Fair: Moderately Clean
6-9	Poor: Somewhat Polluted-Usually indicates that organic matter present and microorganisms are decomposing that waste
100 or more	Very Poor: Very Polluted - Contains organic matter

Chemical oxygen demand (COD) is a slightly better mode used to measure pollution load in water. It is the measure of oxygen equivalent of the requirement of oxidation of total organic matter (i.e. biodegradable and non-biodegradable) present in water.

Major Water Pollutants

1. Industrial waste:

- Industries produce toxic chemicals and pollutants which cause damage to us and our environment.
- They contain pollutants such as lead, mercury, sulphur, asbestos, nitrates and many other harmful chemicals.
- Many industries do not have proper waste management system and drain the waste into rivers, canals and later in to sea.
- The toxic chemicals have the capability to change the color of water, cause Eutrophication, change the temperature of water and pose serious hazard to water organisms.



...Major Water Pollutants

2. Sewage and Wastewater:

- The sewage and waste water that is produced by each household is released into sea with fresh water.
- The sewage water carries harmful bacteria and chemicals that can cause serious health problems.
- Pathogens in sewers are a common water pollutant and thereby spread diseases.

Example: Bacteria- *Vibrio cholera*,
Hepatitis virus, Poliomyelitis virus.



...Major Water Pollutants

3. Mining activities:

→ Mining is the process of crushing the rock and extracting coal and other minerals from underground.

→ These elements when extracted in the raw form contains harmful chemicals and can increase the amount of toxic elements when mixed up with water which may result in health problems.

→ Mining activities emit several metal waste and sulphides from the rocks and is harmful for the water.

Example: Coal Washeries



...Major Water Pollutants

4. Marine dumping:

→ The garbage produced by households in the form of paper, aluminum, rubber, glass, plastic, and food if not segregated and recycled will result in being dumped into the sea.

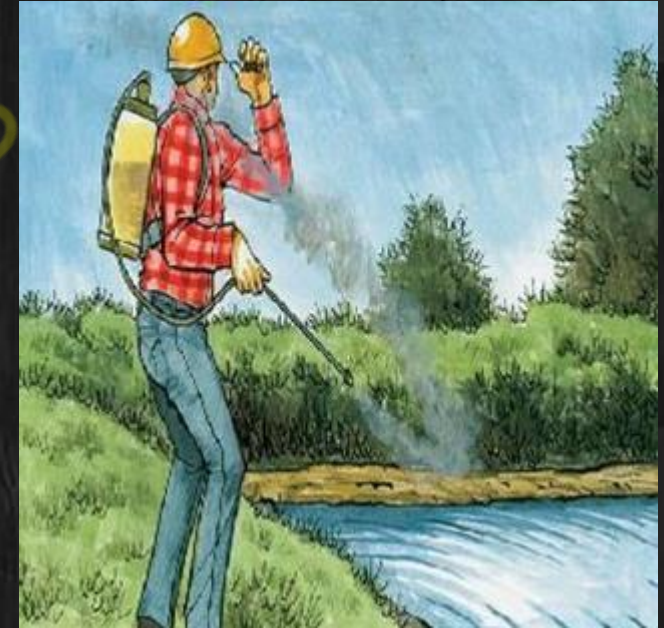
→ These items take from 2 weeks to 200 years to decompose. When such items enters the sea, they not only cause water pollution but also harm animals in the sea.



...Major Water Pollutants

5. Chemical fertilizers and pesticides:

- When it rains, the chemicals mix up with rainwater and flow down into rivers and canals which pose serious damages for aquatic animals.
- Eutrophication and Pesticide contamination of seas.
- Marine Dead Zones



...Major Water Pollutants

6. Thermal Pollution:

- Power plants - thermal and nuclear, chemical and other industries use a lot of water for cooling purposes, and the used hot water is discharged into rivers, streams or oceans.
- Discharge of hot water may increase the temperature of the receiving water by 10 to 15 °C above the ambient water temperature. This is thermal pollution.
- Increase in water temperature decreases dissolved oxygen in the water, also the BOD is increased as metabolism fastens in hot water.
- A sudden rise in temperature kills fishes and other aquatic animals also because of thermal shock.



...Major Water Pollutants

7. Radiation Pollution:

- Nuclear accidents near water bodies or during natural calamities like tsunami and earthquakes pose the risk of radiation leakage (radiation exposure) into water bodies. E.g. Fukushima Daiichi nuclear disaster.
- Radiation exposure causes mutations in the DNA of marine organisms. If those mutations are not repaired, the cell may turn cancerous.
- Radioactive iodine tends to be absorbed by the thyroid gland and can cause thyroid cancer.



...Major Water Pollutants

8. Oil Spills:

- Oil being lighter than water covers the water surface as a thin film, cutting off oxygen to floating plants and other producers.
- Within hours of an oil spill, the fishes, shellfish, plankton die due to suffocation and metabolic disorders.
- Birds and sea mammals that consume dead fishes and plankton die due to poisoning.



...Major Water Pollutants

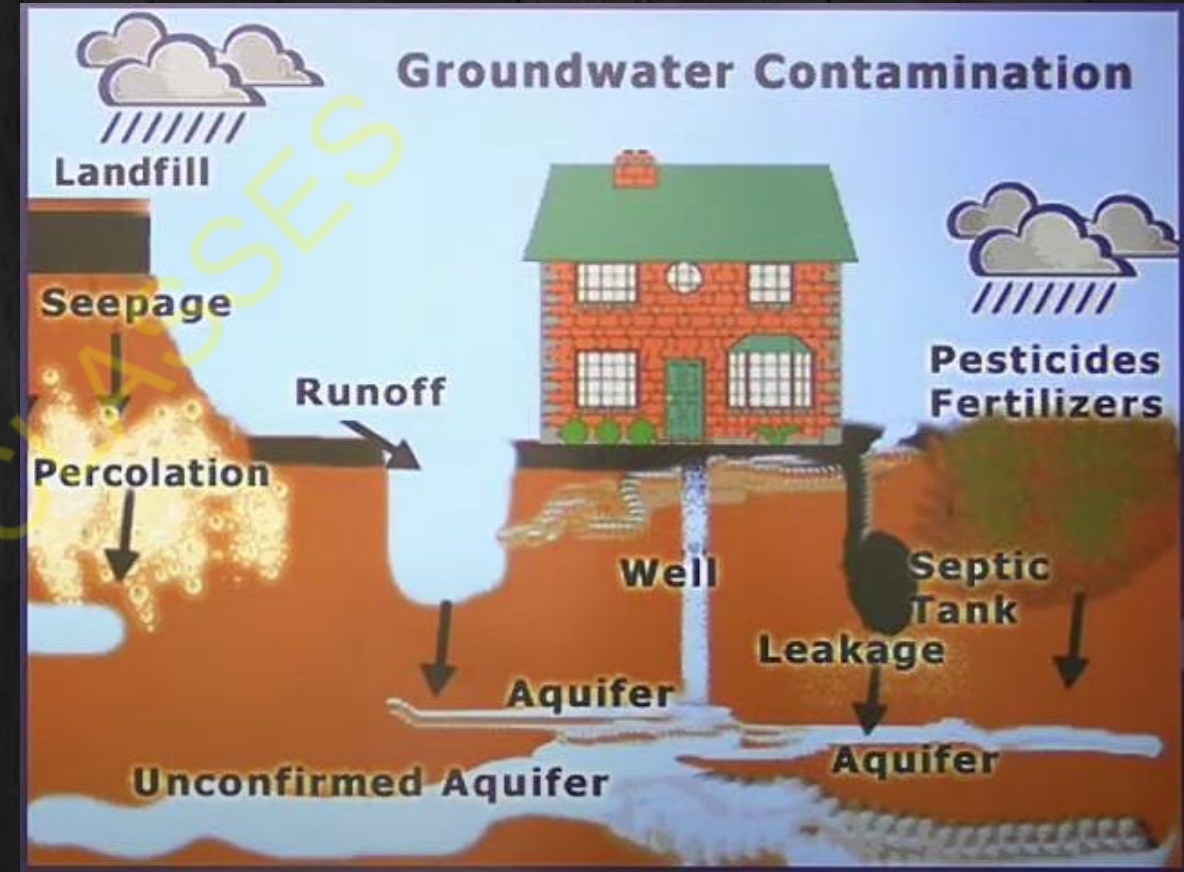
9. Underground water pollution:

→ Seepage from industrial and municipal wastes and effluents, sewage channels and agricultural runoff pollute the groundwater

→ Pollutants like fluorides, uranium, heavy metals and nutrients like nitrates and phosphates are common in many parts of India.

→ **Leakages in:**

- ✓ Landfills
- ✓ Septic Systems
- ✓ Storage Tanks



Effects Of Water Pollution

Effects on aquatic ecosystem

- Polluted water reduces Dissolved Oxygen (DO) content, thereby, eliminates sensitive organisms like plankton, molluscs and fish etc.
- Eutrophication

Effects on human health:

- The polluted water usually contains pathogens like virus, bacteria, parasitic protozoa and worms, therefore, it is a source of water borne diseases like jaundice, cholera, typhoid, amoebiasis etc.
- Mercury compounds in waste water are converted by bacterial action into extremely toxic methyl mercury, which can cause numbness of limbs, lips and tongue, deafness, blurring of vision and mental derangement. (Minamata)

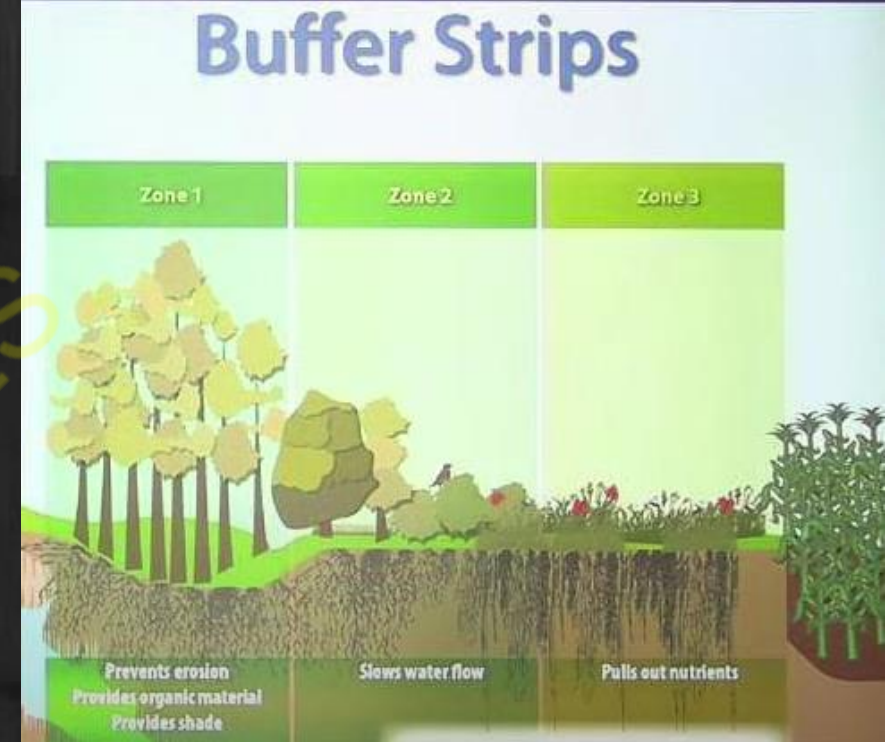
- Water contaminated with cadmium can cause **itai itai disease** also called **ouch-ouch** disease (a painful disease of bones and joints) and cancer of lungs and liver.
- The compounds of lead cause anaemia, headache, loss of muscle power and bluish line around the gum.

Biomagnification

- When the pollutant is persistent organic pollutant like DDT, Mercury

Control Measures Against Water Pollution

1. Riparian buffers
2. Treatment of sewage water and the industrial effluents should be done before releasing it into water bodies.
3. Hot water should be cooled before release from the power plants
4. Domestic cleaning in tanks, streams and rivers, which supply drinking water, should be prohibited.
5. Excessive use of fertilizers and pesticides should be avoided.
6. Organic farming and efficient use of animal residues as fertilizers.



Eutrophication

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→ Eutrophication is characterized by excessive plant and algal growth due to the increased availability of one or more limiting growth factors needed for photosynthesis, such as the:

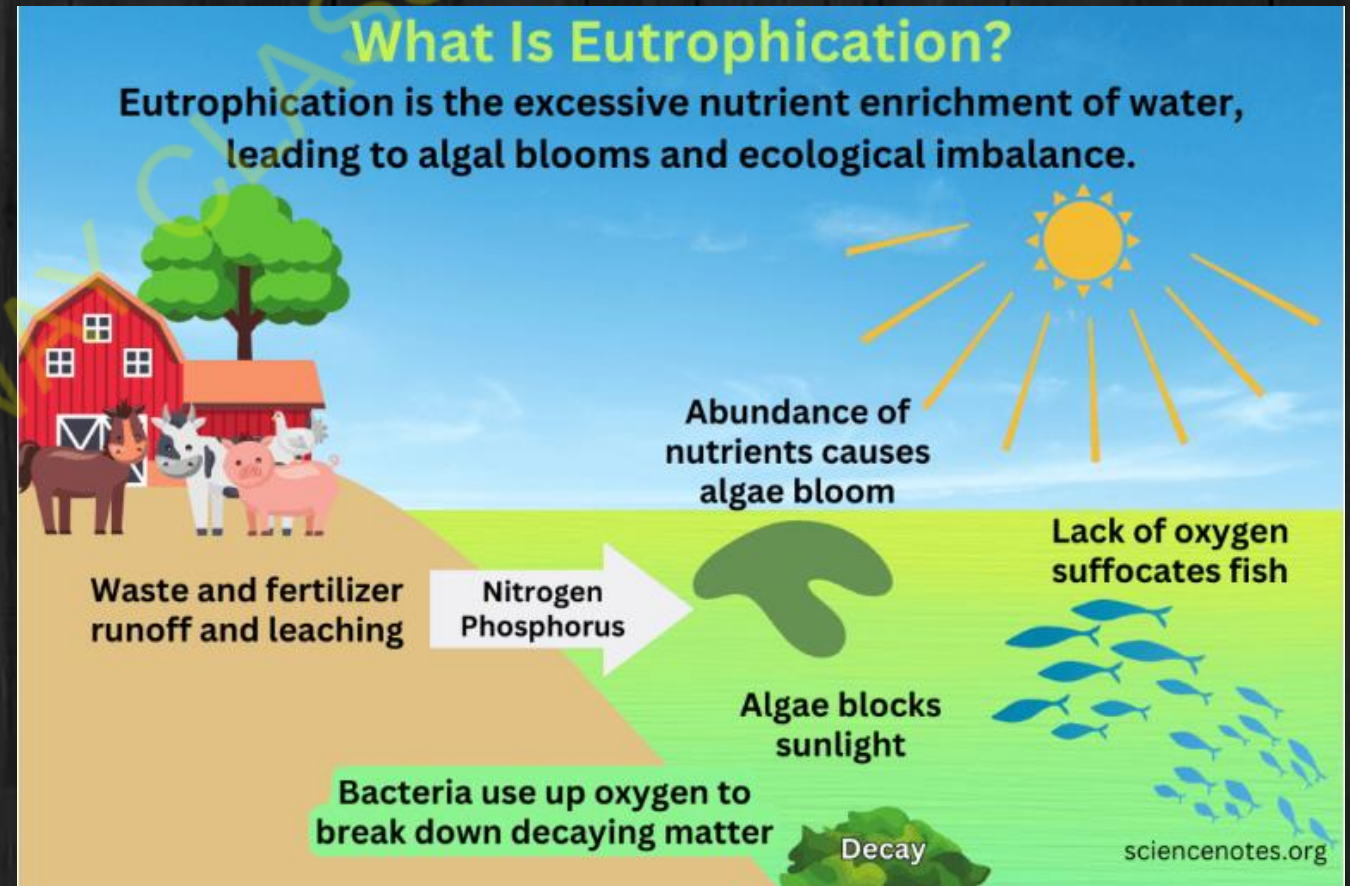
- ✓ Sunlight
- ✓ Carbon dioxide
- ✓ Nutrient fertilizers.

→ Eutrophication occurs naturally over centuries as lakes age and are filled in with sediments.

→ However, human activities have accelerated the rate and extent of eutrophication through both point-source discharges and non-point loadings of limiting nutrients, such as nitrogen and phosphorus, into aquatic ecosystems (i.e., cultural eutrophication).



- 1. Nutrient load up:** excessive nutrients from fertilizers are flushed from the land into rivers or lakes by rainwater.
- 2. Plants flourish:** these pollutants cause aquatic plant growth of algae, duckweed and other plants.
- 3. Algae blooms, oxygen is depleted:** algae blooms, preventing sunlight reaching other plants. The plants die and oxygen in the water is depleted.
- 4. Decomposition further depletes oxygen:** dead plants are broken down by bacteria decomposers, using up even more oxygen in the water.
- 5. Death of the ecosystem:** oxygen levels reach a point where no life is possible. Fish and other organisms die .



Dead Zones :

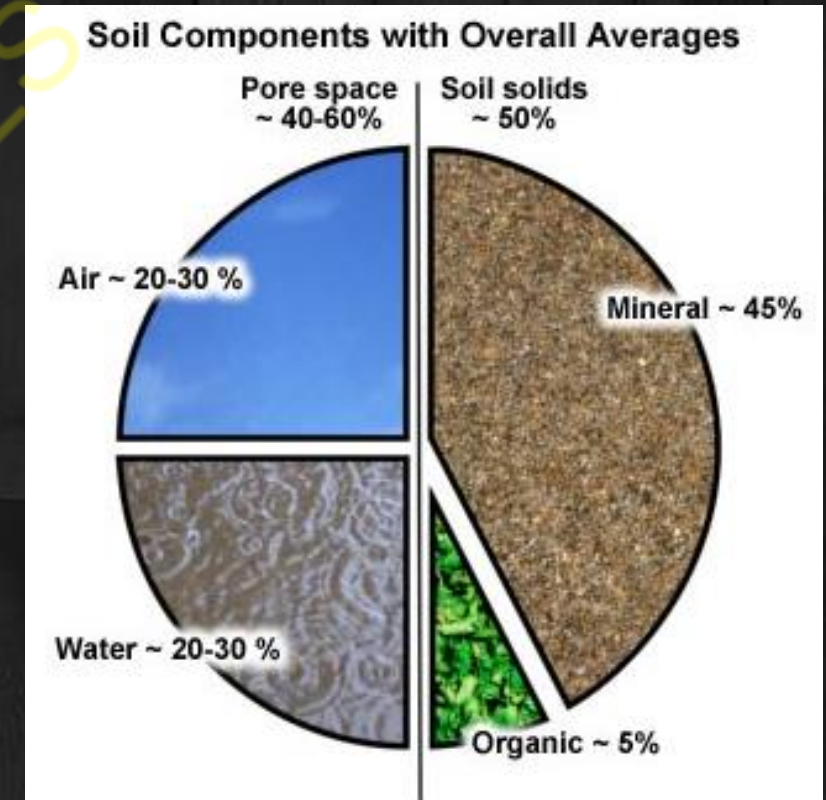
- "Dead Zones" are regions where oxygen concentrations are very low. It may occur in both fresh and marine ecosystem.
- This condition of oxygen deficiency, known as hypoxia. (oxygen level < 2mg/lit)
- In the absence of sufficient oxygen, animals and plants either die or leave producing the dead zone.

Impacts Of Eutrophication

- Threatens the survival of fish and other aquatic life forms.
- Deterioration of water quality and limits access to safe drinking water
- Poisoning and impact on human health
- Endangers fishing
- Degradation of recreational opportunities

Soil Pollution

- Soil is a thin layer of organic and inorganic materials that covers the Earth's rocky surface.
- “Soil pollution is defined as the 'addition of substances to the soil, which adversely affects physical, chemical and biological properties of soil and reduces its productivity”.
- It is a build-up of persistent toxic compounds, chemicals, salts, radioactive materials, or disease-causing agents in the soil which have adverse effects on plant growth, human and animal health.



Sources of Soil Pollution

1. Industrial Wastes:

→ Industrial waste includes chemicals such as mercury, lead, copper, zinc, cadmium, cyanides, thiocyanates, chromates, acids, alkalies, organic substances, etc.

2. Pesticides:

→ Pesticides are chemicals that include insecticides, fungicides, algicides, rodenticides, weedicides sprayed in order to improve productivity of agriculture.

3. Fertilizers :

→ Chemical fertilizers are added to the soil for increasing crop yield.

→ Excess use of chemical fertilizers reduces the population of soil-borne organisms and the crumb structure of the soil, productivity of the soil and increases salt content of the soil.

...Sources of Soil Pollution

4. Discarded materials:

→ It includes concrete, asphalt, rungs, leather, cans, plastics, glass, discarded food, paper and carcasses.

5. Radioactive wastes:

→ Radioactive elements from mining and nuclear power plants, find their way into water and then into the soil.

6. Other pollutants:

→ Many air pollutants (acid rain) and water pollutants ultimately become part of the soil.

Effects of soil pollution

Agriculture

- Reduced soil fertility
- Reduced nitrogen fixation
- Increased erosion
- Loss of soil and nutrients
- Reduced crop yield
- Increased salinity

Health

- Dangerous chemicals polluting underground water
- Bio-magnification
- Release of pollutant gases
- Release of radioactive rays causing health problems

Environment

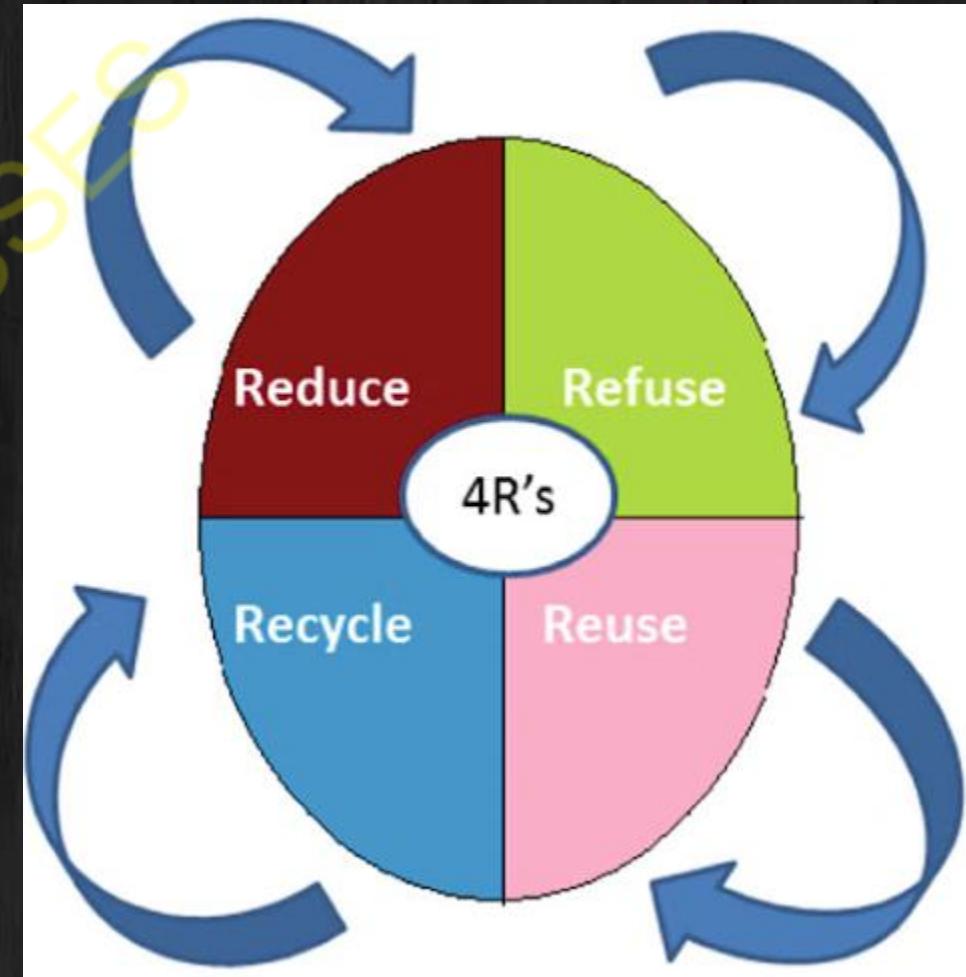
- Reduced vegetation
- Ecological imbalance
- Imbalance in soil fauna and flora

Urban Areas

- Clogging of drains
- Inundation of areas
- Foul smell and release of gases
- Waste management problems

Control measures against Soil Pollution

- **Crop rotation and organic farming**
- Reducing chemical fertilizer and pesticide use
- Use of bio pesticides, bio fertilizers
- Organic farming
- Afforestation and Reforestation
- Solid waste treatment
- Reduction of waste from construction areas
- Four R's: Refuse, Reduce, Reuse, and Recycle



Noise Pollution

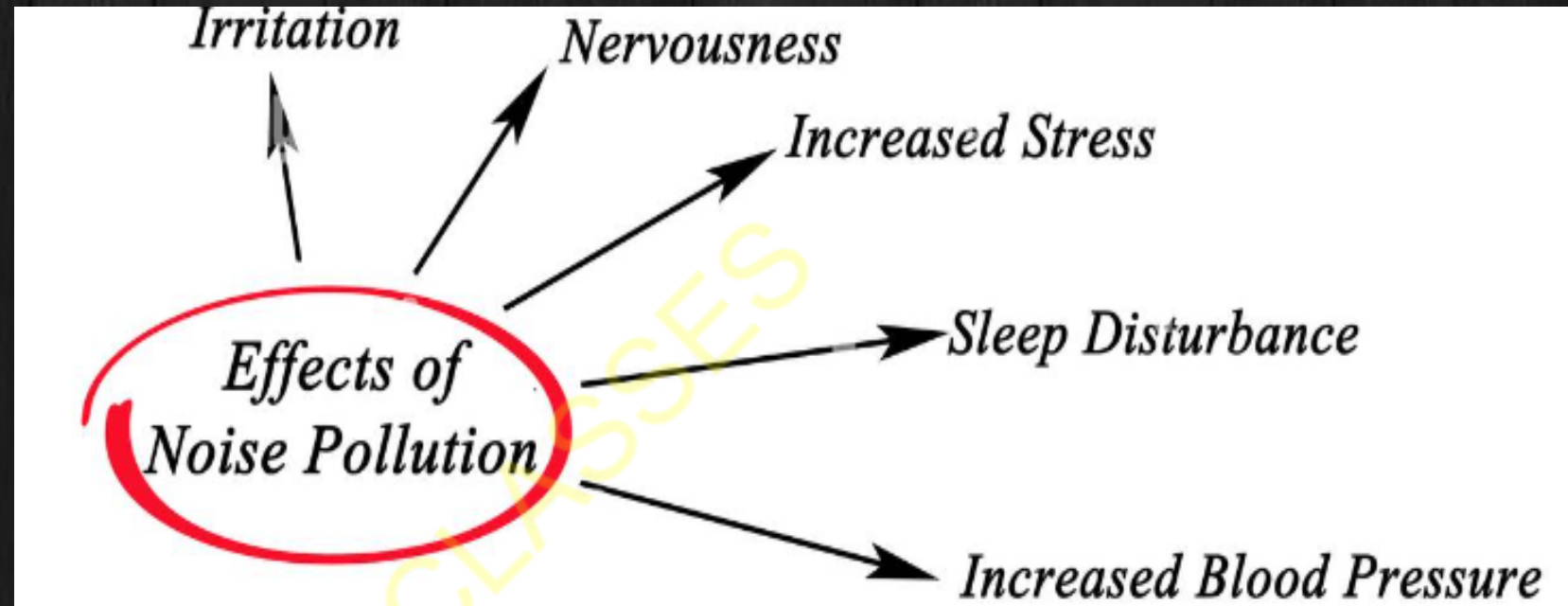
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- Noise pollution is an unpleasant noise created by people or machines that can be annoying, distracting, intrusive, and/or physically painful.
- Noise pollution comes from sources such as "road traffic, jet planes, garbage trucks, construction equipment, manufacturing processes, leaf blowers.
- Sound is measured in decibels (dB).
- A person's hearing can be damaged if exposed to noise levels over 75 dB over a prolonged period of time.
- The World Health Organization recommends that the sound level indoors should be less than 30 dB.

NOISE LIMITS		
Category of Area	Zone Limits in dB (A)	
	Day (6 am-10 pm)	Night (6 am-10 am)
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence Zone	50	40

Impacts of Noise Pollution

Annoyance: It creates annoyance to the receptors due to sound level fluctuations. The a- periodic sound due to its irregular occurrences causes displeasure to hearing and causes annoyance.



Physiological effects: The physiological features like breathing amplitude, blood pressure, heart- beat rate, pulse rate, and blood cholesterol are affected.

Loss of hearing: Long exposure to high sound levels causes loss of hearing. This is mostly unnoticed but has an adverse impact on hearing function.

Human performance: The working performance of workers/humans will be affected as it distracts the concentration.

How to Prevent Noise Pollution

Use Noise Barriers: Install soundproof windows and doors to block external noise.

Limit Use of Loud Appliances: Reduce the use of loud machinery and appliances, especially during nighttime.

Maintain Vehicles and Machinery: Ensure regular maintenance of vehicles and industrial machinery to minimize noise emissions.

Implement Quiet Zones: Establish quiet zones in residential areas and near schools and hospitals.

Plant Trees and Shrubs: Vegetation can act as a natural sound barrier, reducing noise pollution.

Public Awareness: Educate the public about the effects of noise pollution and ways to reduce it.

Regulations and Policies: Enforce stricter noise control regulations and encourage compliance.

Source	Sound Intensity Level SIL (dB)
Jet engine at 10 m	150
Jet engine	140
Jack hammer	130
Car horn	120 (pain threshold)
Rock band	110
Machine shop	100
Train	90
Vacuum cleaner	80
TV	70
Conversation	60
Office	50
Library	40
Hospital	30
Broadcast studio	20
Rustle of leaves	10
Threshold of hearing	0



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Solid Wastes

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- Solid waste is the **unwanted** or **useless** solid materials generated from human activities in residential, industrial, or commercial areas.
- It may be categorized in three ways:
 - ✓ **Origin** (domestic, industrial, commercial, construction or institutional)
 - ✓ **Contents** (organic material, glass, metal, plastic paper, etc)
 - ✓ **Hazard potential** (toxic, non-toxin, flammable, radioactive, infectious, etc)



Types of Solid Waste

→ It can be classified into different types depending on their source:

1. Municipal Solid Waste (MSW)
2. Industrial Solid Waste (ISW)
3. Biomedical waste or hospital waste

→ Open-burning reduces the volume of the wastes

→ Open dumps often serve as the breeding ground for rats and flies.

Effects of Plastic Waste

→ Conventional plastics have been associated with reproductive problems in both humans and wildlife.

→ **Dioxin** (highly carcinogenic and toxic) by-product of the manufacturing process is one of the chemicals believed to be passed on through breast milk to the nursing infant.

[Dioxin is one of the POP] POP-Persisting Organic Pollutant

→ Burning of plastics, especially PVC releases dioxin and also furan into the atmosphere.

Municipal Solid Waste

- Municipal solid waste consists of household waste, construction and demolition debris, sanitation residue, and waste from streets.
- More than 25% of the municipal solid waste is not collected at all.
- 70% of the Indian cities lack adequate capacity to transport it and there are no sanitary landfills to dispose of the waste.
- The existing landfills are neither well equipped nor lined properly to protect against contamination of soil and groundwater.
- Increasing use of cans, aluminium foils, plastics, and other such non-biodegradable items that cause incalculable harm to the environment.

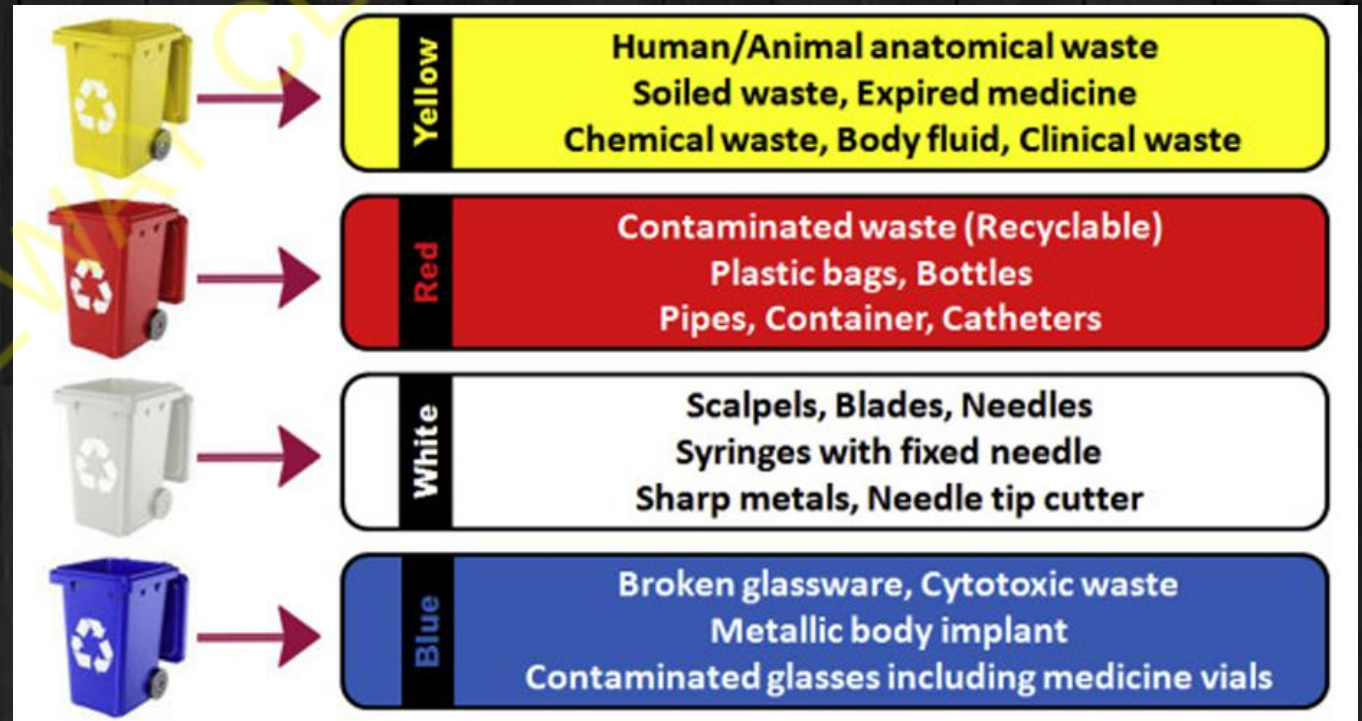


Industrial Solid Waste (ISW) and Hazardous waste

- Industrial and hospital waste is considered **hazardous** as they contain toxic substances.
- Hazardous wastes could be highly toxic to humans, animals, and plants and are corrosive, highly inflammable, or explosive.
- India generates around 7 million tonnes of hazardous wastes every year, most of which is concentrated in **four states: Andhra Pradesh, Bihar, Uttar Pradesh, and Tamil Nadu.**
- Household waste that can be categorized as hazardous waste include old batteries, shoe polish, paint tins, old medicines, and medicine bottles.
- In the industrial sector, the major generators of hazardous waste are the metal, chemical, paper, pesticide, dye, refining, and rubber goods industries.
- Direct exposure to chemicals in hazardous waste such as mercury and cyanide can be fatal.

Hospital waste or Biomedical Waste

- Hospital waste is generated during the diagnosis, treatment, of human beings or animals or in research activities, etc.
- These chemicals include formaldehyde and phenols, which are used as disinfectants, and mercury, which is used in thermometers or equipment that measure blood pressure.
- It may include wastes like soiled waste, disposables, anatomical waste, cultures, discarded medicines, chemical wastes, disposable syringes, swabs, bandages, body fluids, human excreta, etc.



Difference between Hazardous Waste and Non-Hazardous Waste

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Hazardous Waste:

- Dangerous to public health or the environment.
- Contains harmful chemicals that can cause health problems
- Can easily catch fire
- Can cause explosions or release toxic gases.
- Can corrode metals and damage skin.

Examples:

- Industrial chemicals
- Pesticides
- Medical waste
- Batteries
- Paints and solvents

Non-Hazardous Waste:

- Not Dangerous to public health or the environment

Inert

- Does not react chemically with other substances.
- Does not contain harmful chemicals.
- Does not easily catch fire.
- Does not corrode materials or damage skin.

Examples:

- Food waste
- Paper and cardboard
- Garden waste

Solid Waste Management

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Treatment and disposal of SW- Open dumps

- Open dumps refer to uncovered areas that are used to dump solid waste of all kinds.
- The waste is untreated, uncovered, and not segregated. It is the breeding ground for flies, rats, and other insects that spread disease.
- The rainwater run-off from these dumps contaminates nearby land and water thereby spreading disease. Treatment by open dumps is to be phased out.

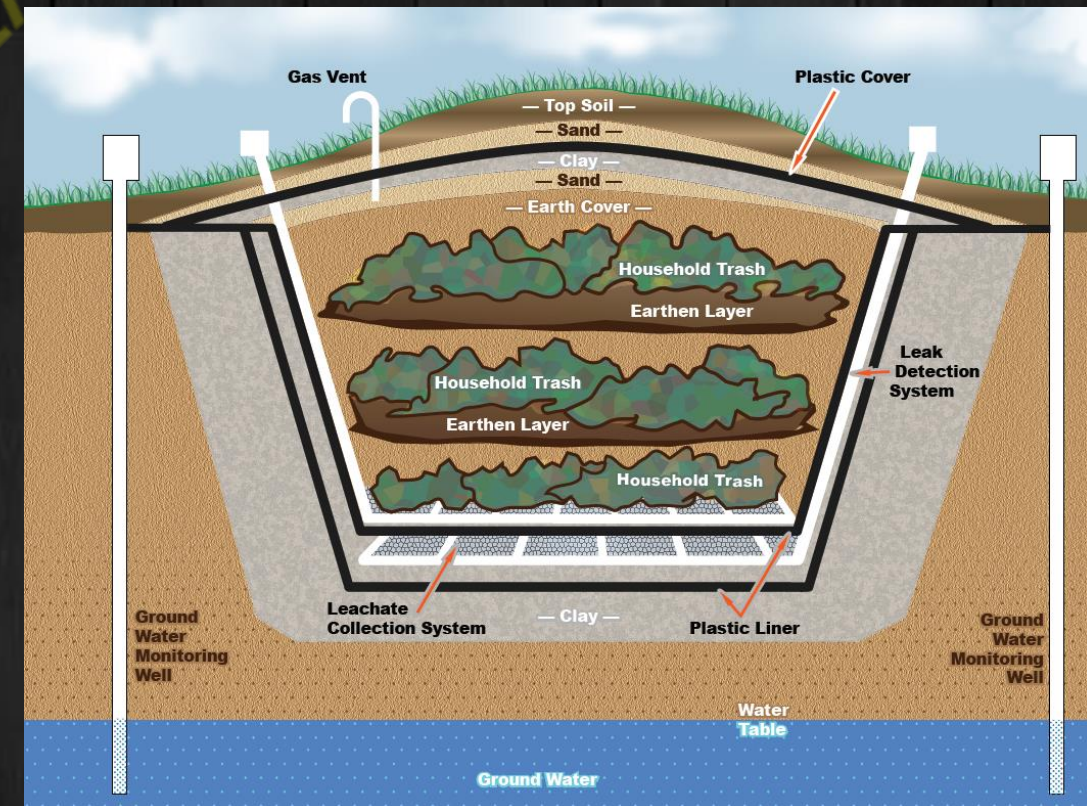


Treatment and disposal of solid waste- Landfills

→ Landfills are generally located in urban areas. It is a pit that is dug in the ground. The garbage is dumped and the pit is covered with soil everyday thus preventing the breeding of flies and rats. Thus, every day, garbage is dumped and sealed.

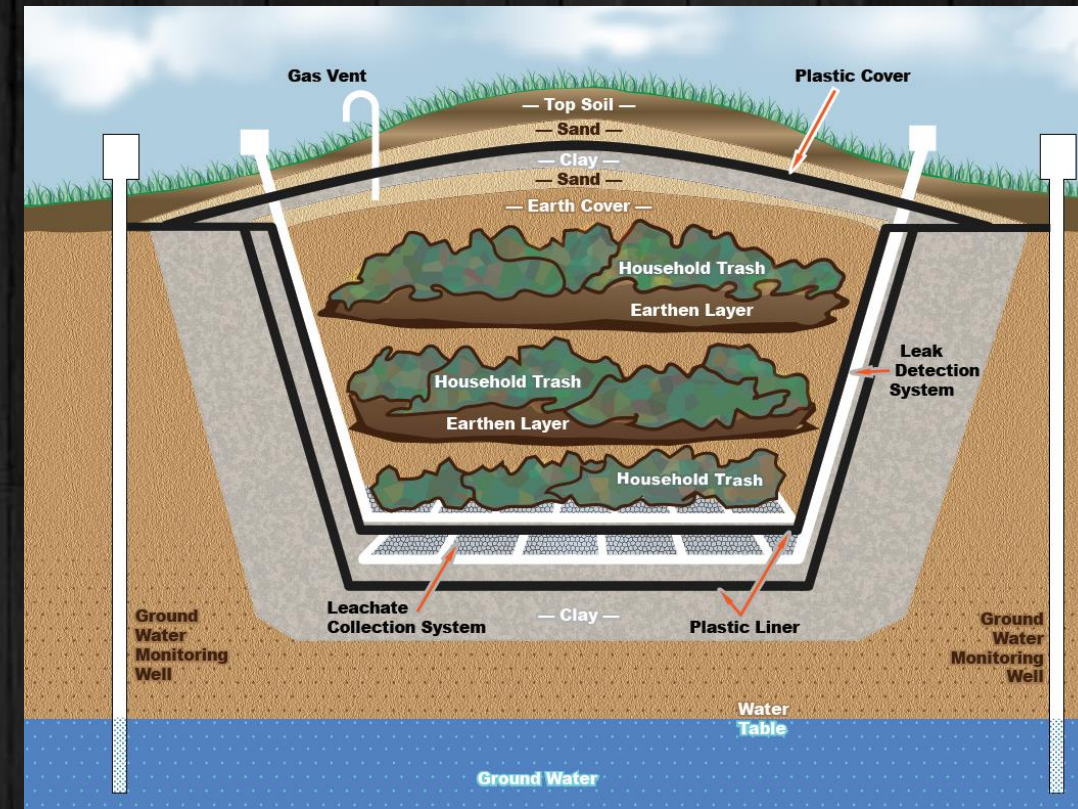
→ After the landfill is full, the area is covered with a thick layer of mud and the site can thereafter be developed as a parking lot or a park.

Problems - All types of waste are dumped in landfills and when water seeps through them it gets contaminated and in turn pollutes the soil and ground water. This contamination of groundwater and soil through landfills is known as **leaching**. It is also prone to fire.



Treatment and disposal of SW- Sanitary landfills

- Sanitary landfills is more hygienic and built in a methodical manner to solve the problem of leaching.
- These are lined with materials that are impermeable such as plastics and clay, and are also built over impermeable soil. Constructing sanitary landfills is costly.



Treatment and disposal of SW- Incineration plants

- The process of burning waste in large furnaces at high temperatures in the presence of oxygen is known as incineration.
- In these plants the recyclable material is segregated and the rest of the material is burnt.
- Waste is converted into carbon dioxide, water-vapour, and ash. Also called the Waste to Energy (WtE) method, it is a means of recovering energy from waste.

It's advantages include:

- waste volume reduction,
- Cutback on transportation costs and
- Reduction of greenhouse gas emissions.
- However, when garbage is burned, pollutants, such as mercury, lead, dioxins may be released into the atmosphere, and cause health issues.

Treatment and disposal of solid waste- Pyrolysis

→ It is a process of combustion in absence of oxygen or the material burnt under controlled atmosphere of oxygen.

→ It is an alternative to incineration. The gas and liquid thus obtained can be used as fuels.

→ Pyrolysis of carbonaceous wastes like firewood, coconut, palm waste, corn combs, cashew shell, rice husk paddy straw and saw dust, yields Syngas, Char and Tar (Pyrolysis Oil)

Biomass type	Char (%)	Syngas (%)	Tar (%)
Ligneous biomass	21	71	8
Agricultural residues	29	49	22
Cultivated biomass	30	60	10

Composition (% vol)	
CH_4	21
CO	29
CO_2	38
H_2	7
N_2	5

Syngas Composition



Char



Treatment and disposal of solid waste- Composting

- Composting is a biological process in which micro-organisms, mainly fungi and bacteria, decompose organic waste into humus like substance in the presence of oxygen.
- This finished product, which looks like soil, is high in carbon, nitrogen and is an excellent medium for growing plants. It also increases the water holding capacity of the soil.
- Apart from being clean, cheap, and safe, composting can significantly reduce the amount of disposable garbage.

Vermiculture

- It is also known as earthworm farming. In this method, Earth worms are added to the compost.
- These worms break the waste and the added excreta of the worms makes● the compost very rich in nutrients.



Current Challenges of Solid Waste Management in India

Rapid Urbanization:

- Due to rapid urbanization has led to an increase in the volume of waste generated.

Insufficient Infrastructure:

- Many areas lack the necessary infrastructure for effective waste collection, transportation, and disposal.

Lack of Public Awareness:

- Public awareness about waste segregation and recycling is generally low.

Inadequate Funding:

- Limited financial resources for waste management projects.

Regulatory Challenges:

- Enforcement of waste management laws and regulations is often weak.

Informal Sector Involvement:

→ A significant portion of waste management is handled by the informal sector, which lacks proper training and resources.

E-Waste Management:

→ The growing volume of electronic waste poses specific challenges due to its hazardous components.

Landfill Overload:

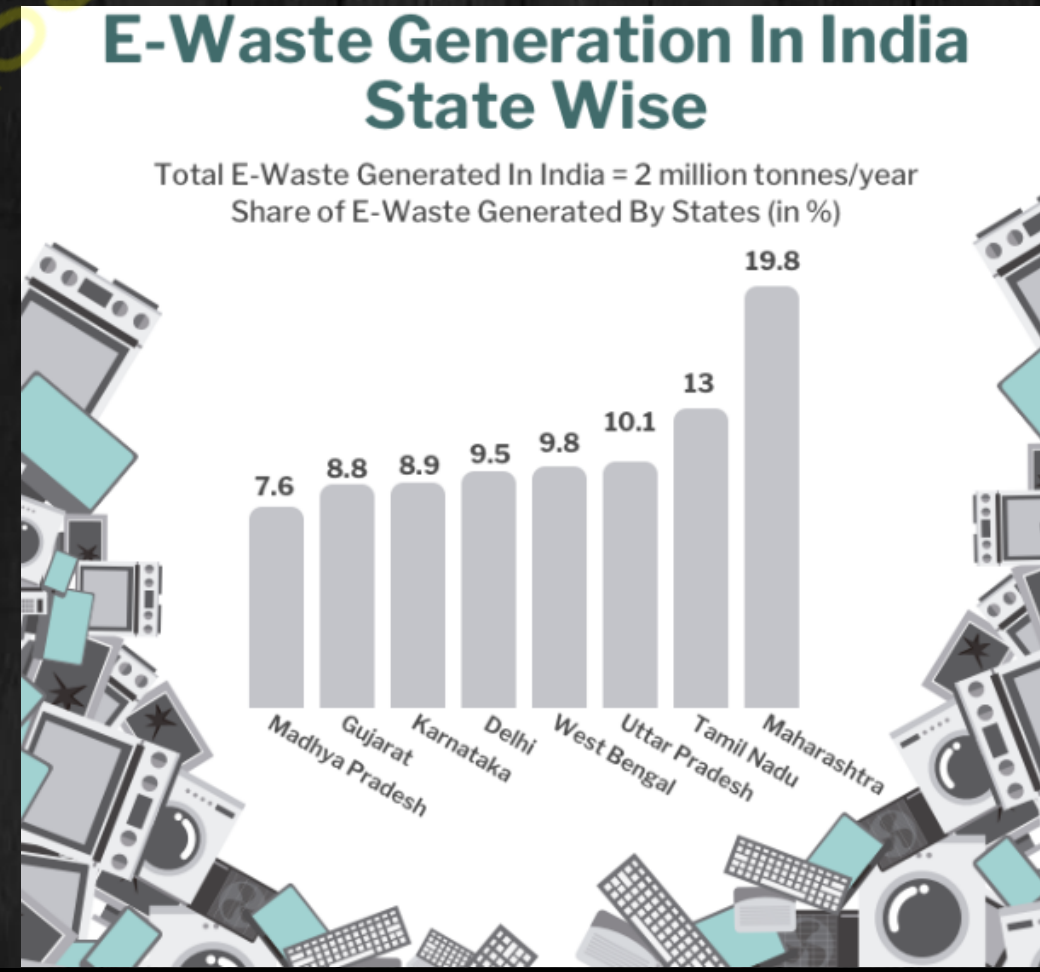
→ Existing landfills are overburdened and often not designed to handle the current volume of waste.

Lack of Data and Monitoring:

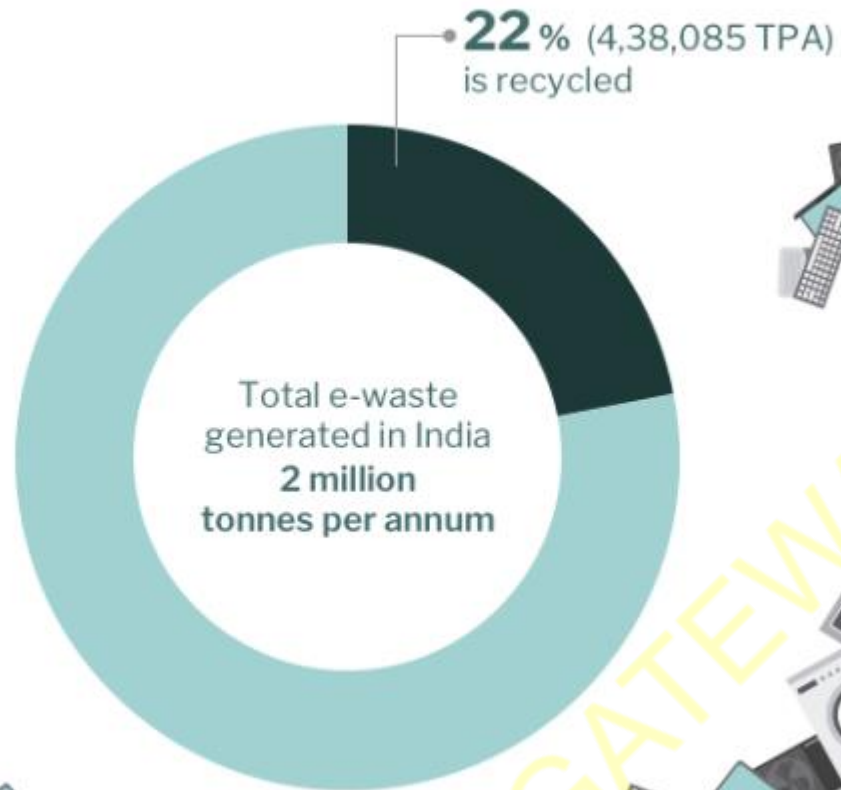
→ Insufficient data collection and monitoring of waste management practices.

E-Waste In India

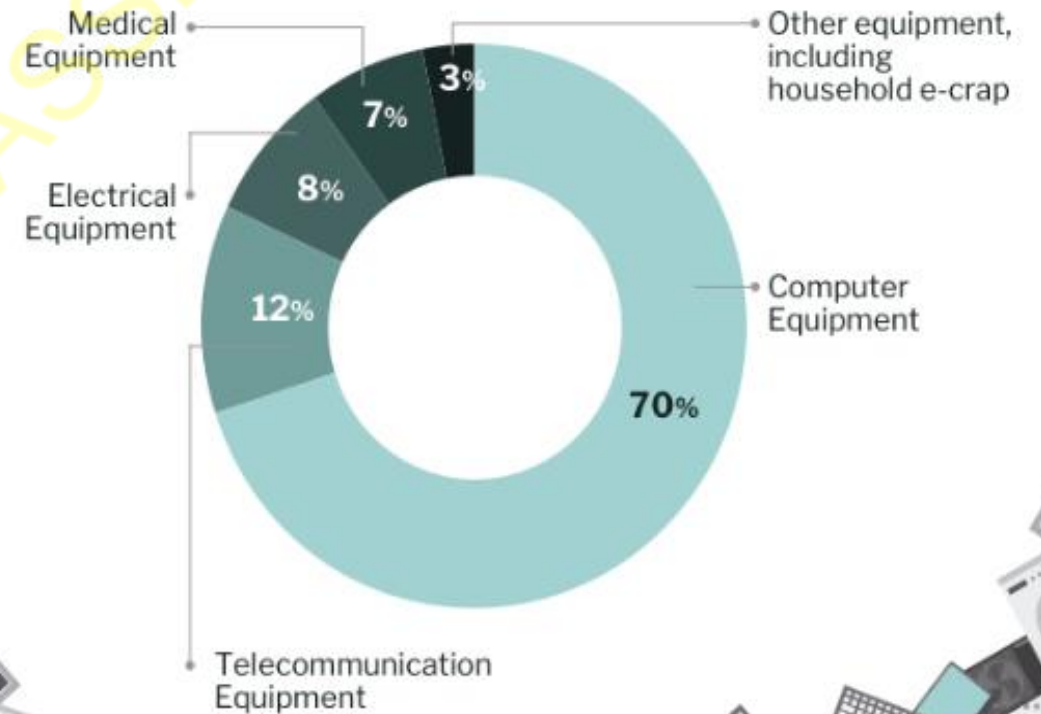
- **The Global E-Waste Monitor 2014**, 1.7 Million tonnes of e-waste generation was reported in the country in 2014.
- Mumbai ranks first in generating e-waste followed by Delhi, Bangalore, Chennai, Kolkata, Ahmadabad, Hyderabad, Pune, Surat and Nagpur.
- The 65 cities generate more than 60% of the total generated e-waste, whereas, 10 states generate 70% of the total e-waste.
- E-Waste Management Rules - 2016 and 2018 Amendment
- Extended Producer Responsibility



E-Waste Recycling In India



Break-up Of E-Waste In India





Thank You